

Experiment to understand feature scaling.



```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.preprocessing import MinMaxScaler
from sklearn.preprocessing import StandardScaler
```

```
data = {
    "Age": [18, 20, 22, 24, 25],
    "Study_Hours": [2, 4, 6, 8, 10],
    "Marks": [40, 50, 65, 80, 95]
}
```

```
df = pd.DataFrame(data)
```

```
print(df)
```

	Age	Study_Hours	Marks
0	18	2	40
1	20	4	50
2	22	6	65
3	24	8	80
4	25	10	95

```
#Min - Max Normalization
scaler = MinMaxScaler()

normalized_data = scaler.fit_transform(df)

normalized_df = pd.DataFrame(
    normalized_data,
    columns=df.columns
)

print(normalized_df.round(2))
```

	Age	Study_Hours	Marks
0	0.00	0.00	0.00
1	0.29	0.25	0.18
2	0.57	0.50	0.45
3	0.86	0.75	0.73
4	1.00	1.00	1.00

```
#Z - score normalization
scaler = StandardScaler()

standardized_data = scaler.fit_transform(df)

standardized_df = pd.DataFrame(
    standardized_data,
    columns=df.columns
)

print(standardized_df.round(2))
```

	Age	Study_Hours	Marks
0	-1.48	-1.41	-1.31
1	-0.70	-0.71	-0.81
2	0.08	0.00	-0.05
3	0.86	0.71	0.71
4	1.25	1.41	1.46